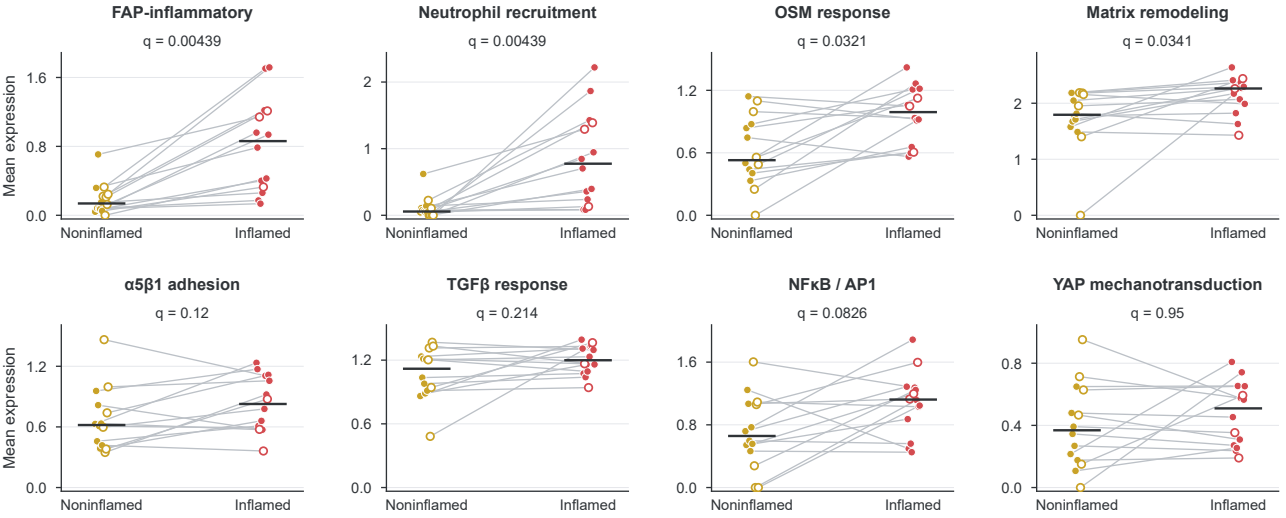


Supplementary Figure 1

A Paired stromal expression of fibroblast-associated programs

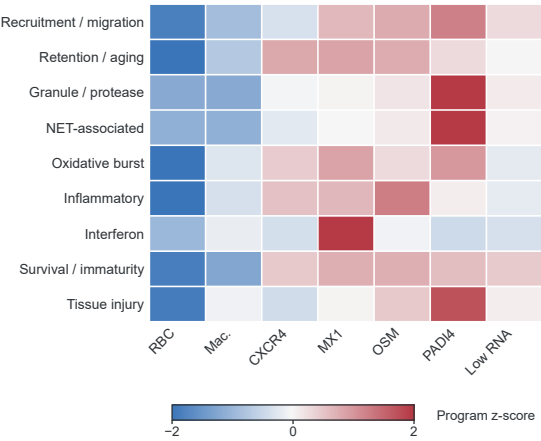
Matched noninflamed and inflamed UC tissue; 14 patients



Paired Wilcoxon q (BH, 8 programs). Lines: patients; bars: medians; open points: <10 recovered cells.

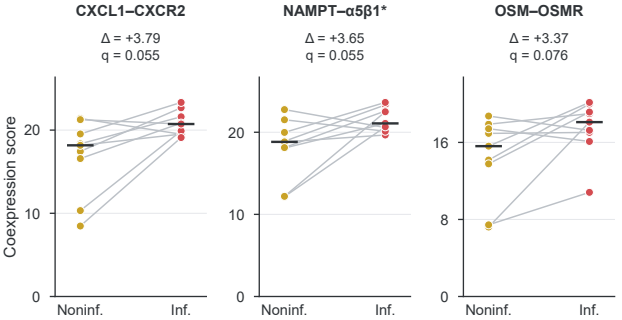
B Pooled RNA programs and annotation controls

All conditions; original gene modules



C Paired ligand–receptor coexpression

9 matched UC patient pairs



Paired Wilcoxon q (BH, 5 original routes).

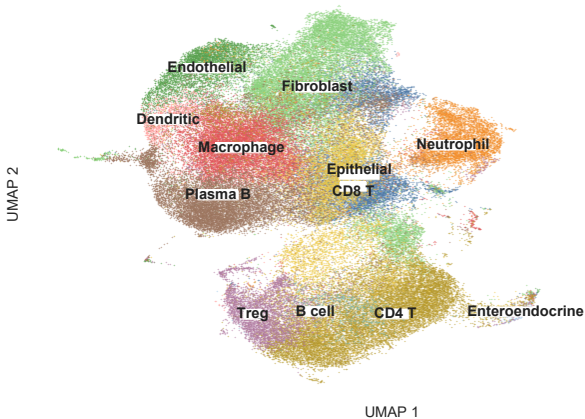
*Receptor summary: min(ITGA5, ITGB1).

Δ : median paired change; lines: patients.

Supplementary Figure 2

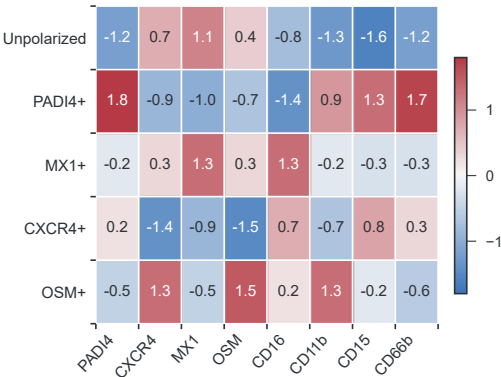
A Broad CODEX cell atlas

245,020 cells; 12 broad annotations



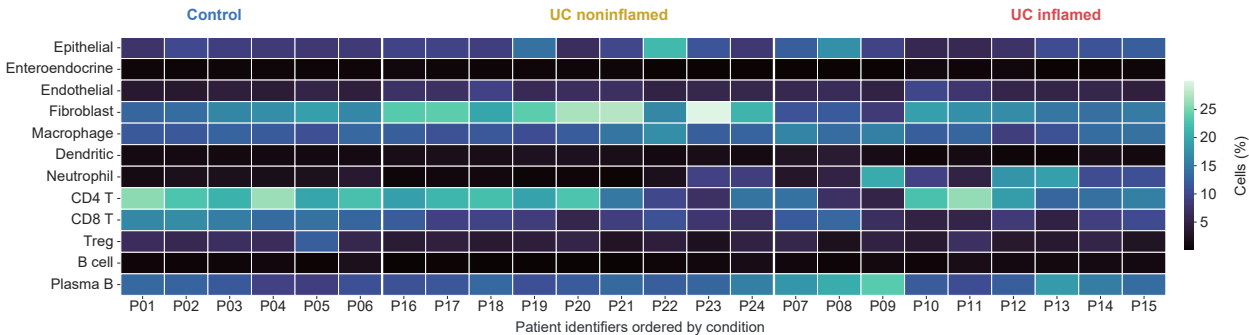
B Neutrophil marker profiles

Median marker expression, z-scored across states



C Patient-level cell composition

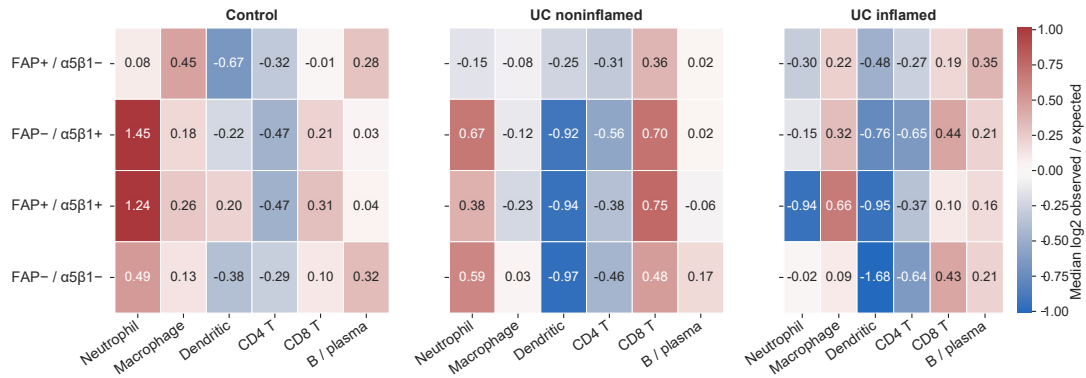
Six control, nine noninflamed UC and nine inflamed UC patients



Supplementary Figure 3

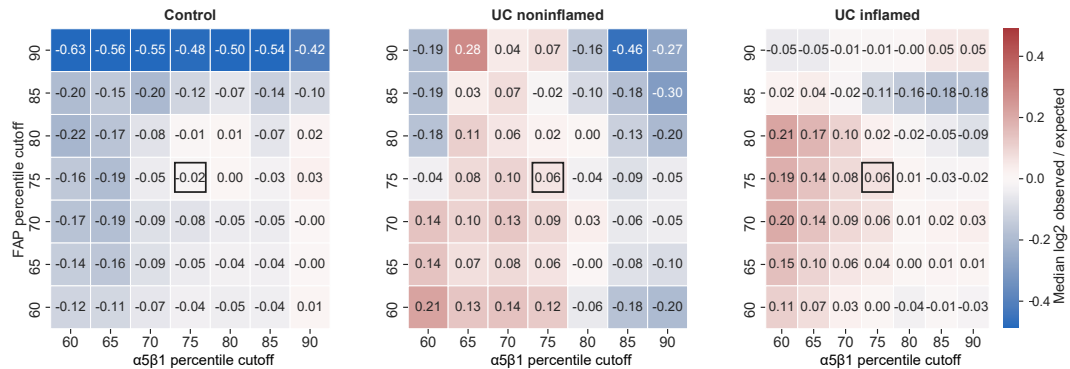
A Immune-lineage spatial specificity

Median log2 observed/expected proximity across FAP/ $\alpha 5\beta 1$ strata



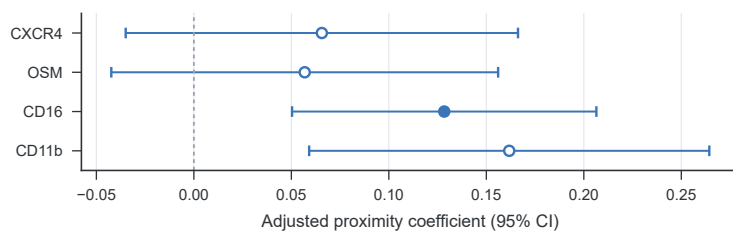
B Threshold sensitivity

FAP and $\alpha 5\beta 1$ percentile cutoffs across tissue conditions



C Adjusted neutrophil proximity effects

Filled point: restricted spatial-null $q < 0.05$; bars show 95% CI

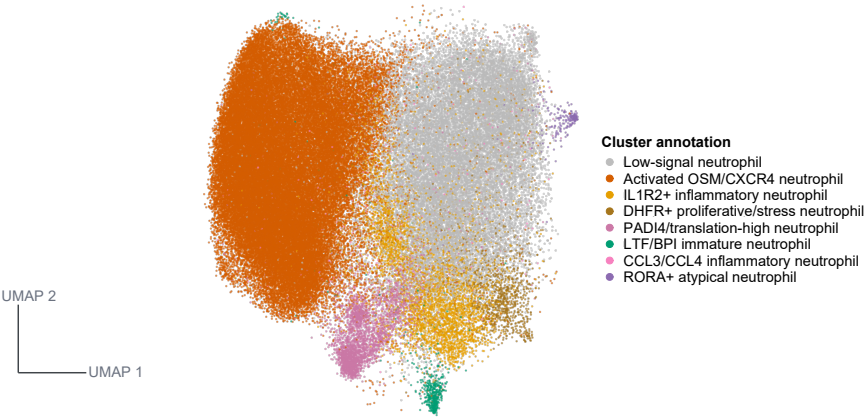


Model q	Spatial-null q
0.242	0.844
0.242	0.843
0.008	0.032
0.008	0.144

Supplementary Figure 5

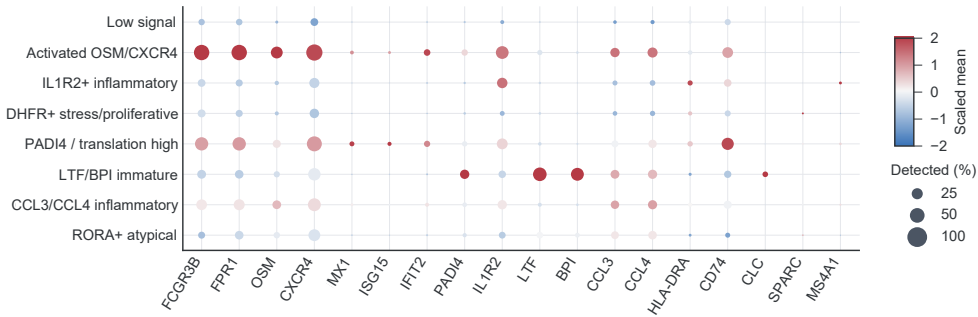
A Neutrophil subcluster atlas

Retained cells colored by transcriptomic subcluster annotation



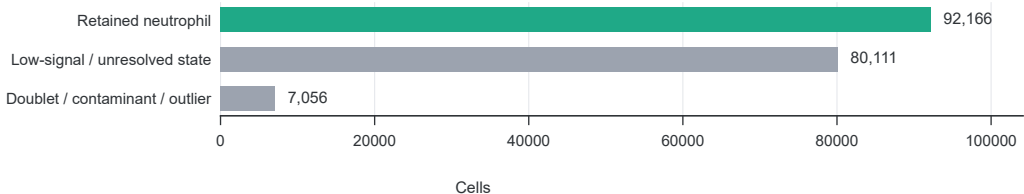
B Marker support for subcluster annotations

Dot size: detected cells; color: scaled mean expression



C Cell-retention and cleanup inventory

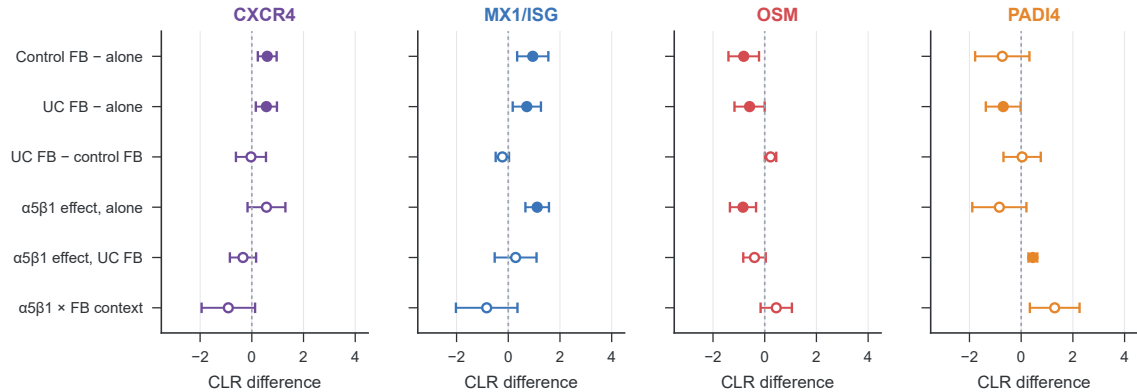
Original counts summarized by disposition



Supplementary Figure 6

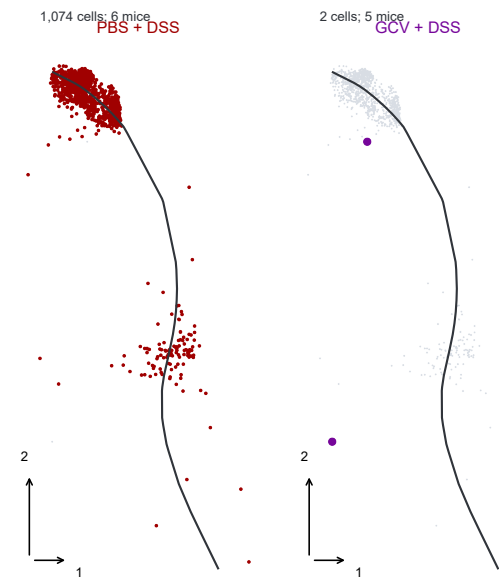
A Recorded-donor neutrophil state contrasts

Six recorded donors; filled points denote BH $q < 0.05$; bars show 95% CI



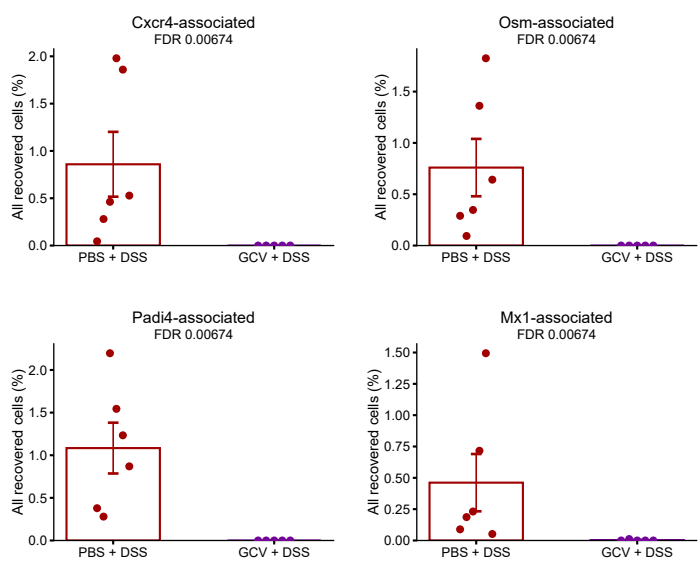
Supplementary Figure 7

A Neutrophil occupancy after FAP ablation

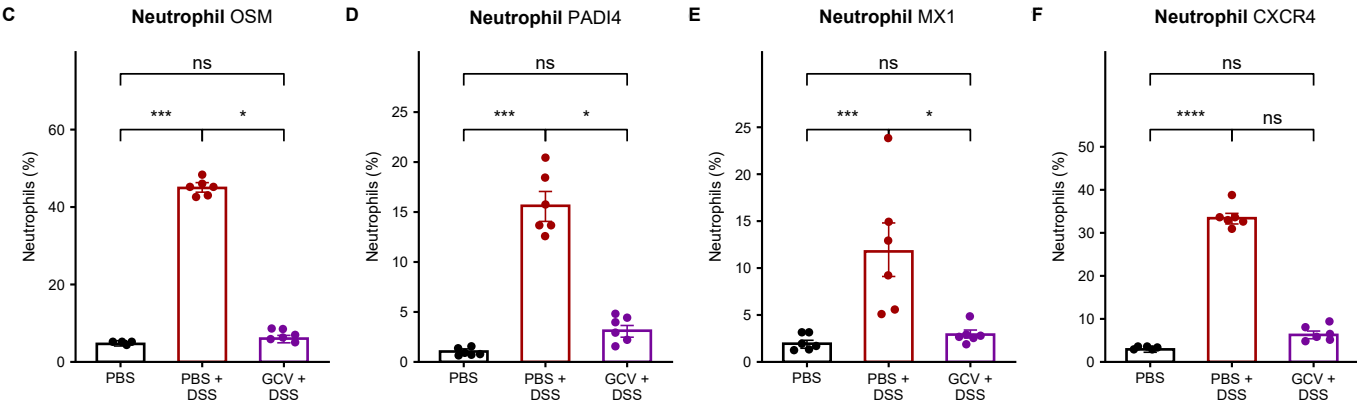


Only two postablation neutrophils: occupancy loss, not a postablation trajectory or within-state estimate.

B Recovered neutrophil-state fractions



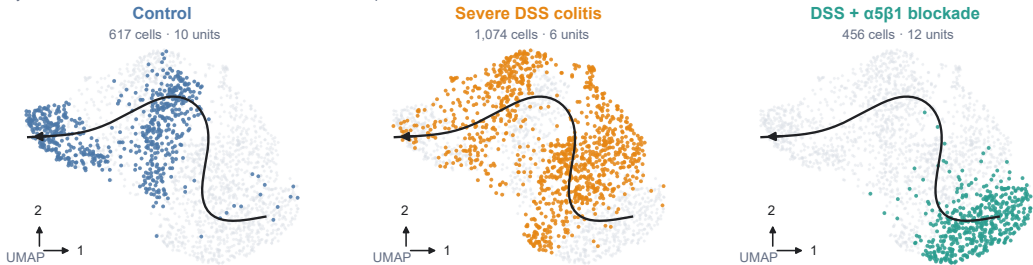
C-F Neutrophil marker readouts supporting Figure 4



Supplementary Figure 8

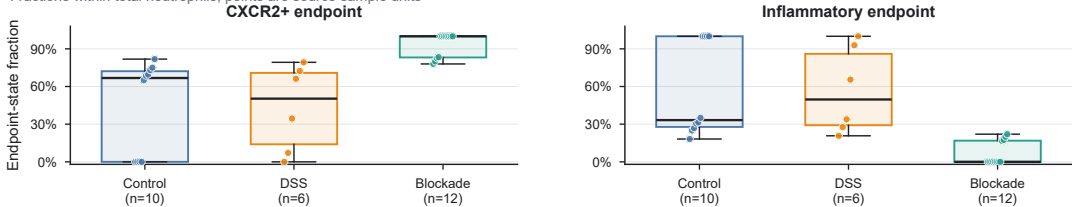
A $\alpha 5 \beta 1$ blockade redistributes occupancy along a shared neutrophil continuum

Grey denotes the full reference manifold; arrow indicates imposed orientation, not measured time



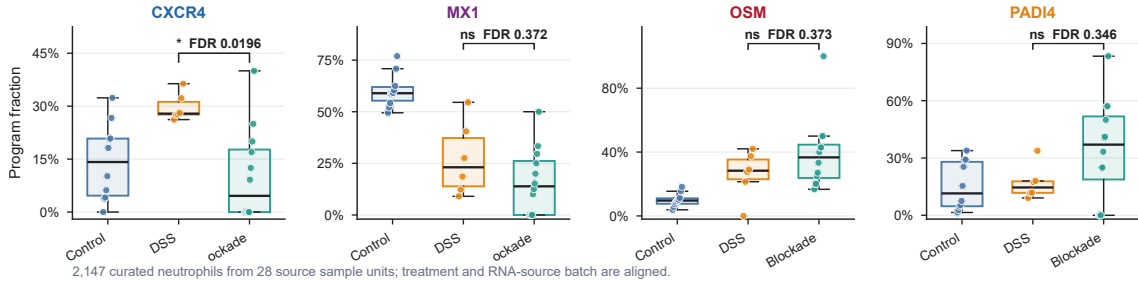
B Endpoint-state redistribution

Fractions within total neutrophils; points are source sample units



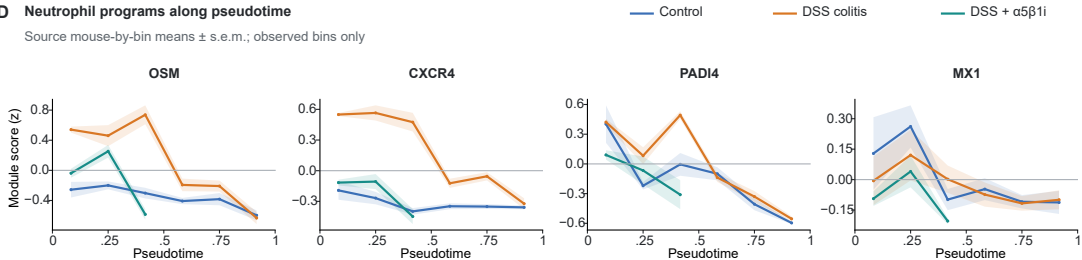
C Transferred human-associated neutrophil programs

Ortholog-mapped dominant program; source DSS-versus-blockade tests retained



D Neutrophil programs along pseudotime

Source mouse-by-bin means $\pm$ s.e.m.; observed bins only



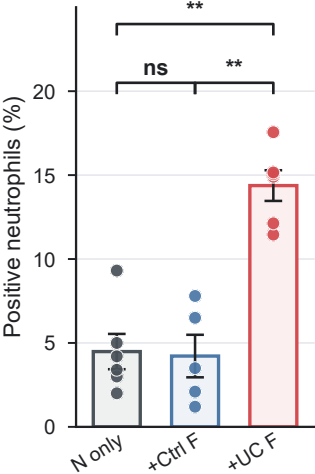
Supplementary Figure 9

Human fibroblast coculture: neutrophil protein readouts

Neutrophils only + control fibroblasts + UC fibroblasts

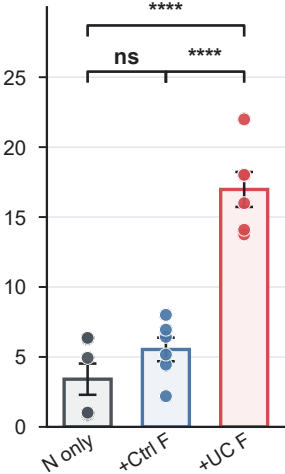
A

PADI4+ neutrophils



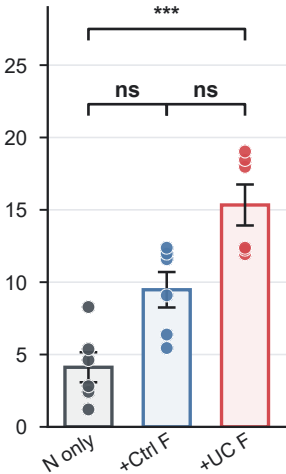
B

OSM+ neutrophils



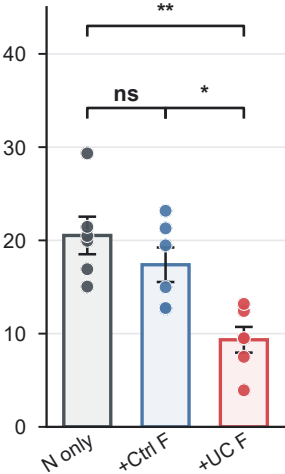
C

MX1+ neutrophils



D

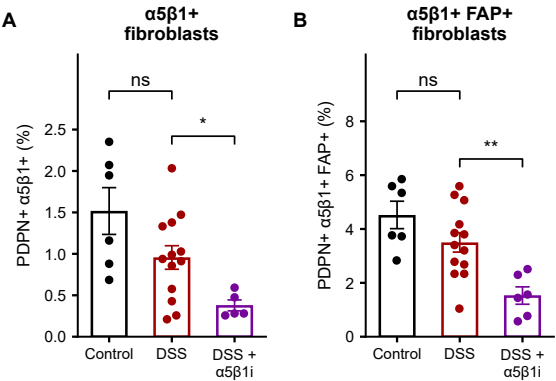
CXCR4+ neutrophils



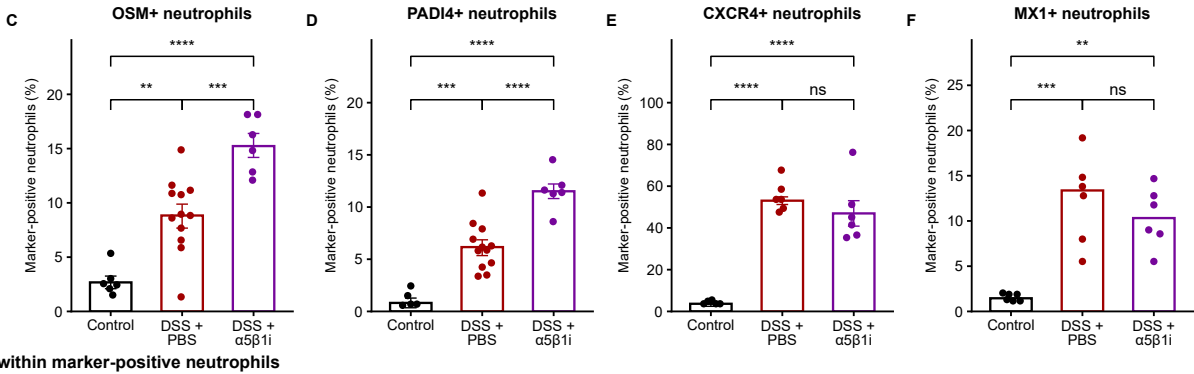
Supplementary Figure 10

DSS colitis: stromal and neutrophil protein readouts after $\alpha 5\beta 1$ blockade

A,B Stromal $\alpha 5\beta 1$ detection



C–F Marker-positive neutrophils



G MPO within marker-positive neutrophils

